

Instructor: Date:

Learner: Regional Branch:

Program: Center:

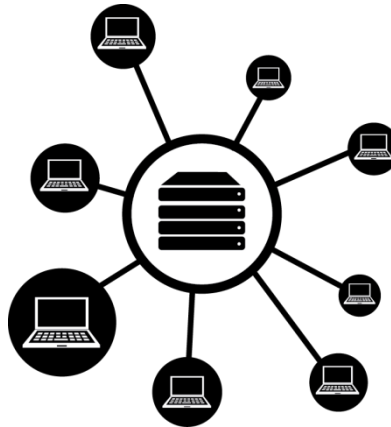
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Let's Start!

A Powerful Way to Structure Large Applications



Learning Outcome:

Build the application installation program, using IT development tools available in the market, based on solution architecture features. Recognizes and implements the different types of architecture styles that can be included in the IT solution.

Can Do Statements:

- I can explain what a message bus architecture is.
- I can describe the advantages of a message bus architecture.
- I can talk about the disadvantages of a message bus architecture.

Ask your partner:

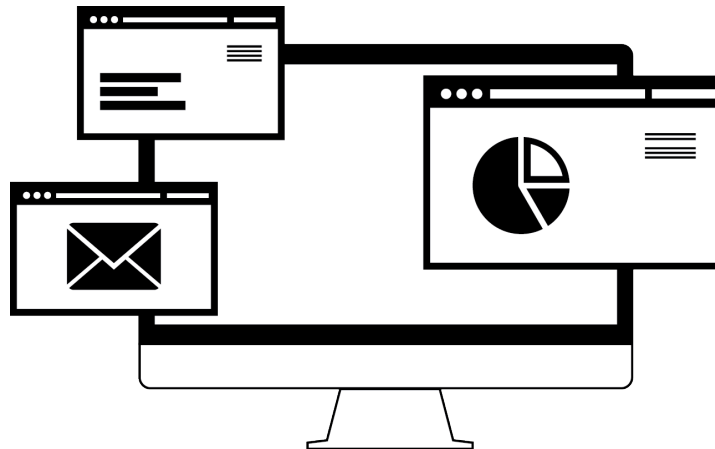
- What are the advantages of working with a software that runs several applications at the same time?
- How secure do you think are systems that send messages indiscriminately to the public?





Let's Explore!

Holly suggests using a message bus architecture.



Instructions: Read the text below.

Holly: You look puzzled. Is everything ok?

Ivan: Well, I am trying to connect several applications, and I need them to communicate with one another. However, if I link all the applications, there will be many point-to-point connections.

Holly: You may need to use a different method if you want to have a less complex connection. Why don't you use a message bus architecture?

Ivan: I've heard of it, but I don't really understand how it works.

Holly: Let me explain to you what a message bus is. It's a virtual machine designed to send messages among different applications. If various components or services need to interact, they will use a message bus. Buses provide reliable messaging capabilities to clients as they bind services from different natures.

Ivan: Is it appropriate for large amounts of data?

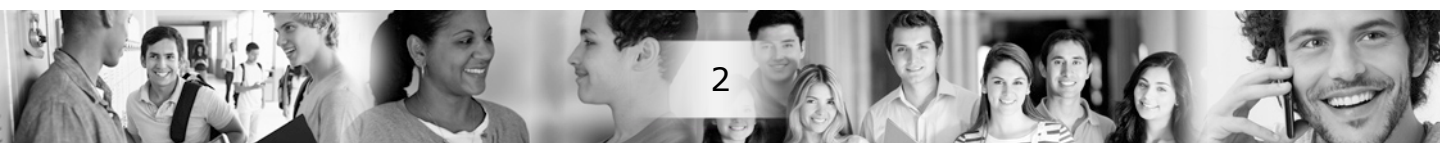
Holly: Sure. A message bus doesn't have a limit in size, and it queues the messages if necessary, but you may use complementary shared memory techniques to manage a bulk of data. To make sure everything is going well with your messaging bus, you just need a console client.

Ivan: What advantages do I get from using this style?

Holly: If you use a message bus, you will reduce complexity in application communication. It also provides modifiability just in case you need to add more components. You don't need to use intermediary programs, and that improves performance. Finally, you may be interested in this style's scalability because new components don't affect the old ones.

Ivan: Are there any disadvantages?

Holly: Well, the communication is a little complex since all parts need to share a protocol. It also has lowered security as the messages are sent indiscriminately. If you need to change the bus interface, modifiability lowers; and finally, it has low tolerance for downtime.





Let's Practice!

Holly will explain what a message bus architecture is.

Instructions: Read the sentences. Write the missing words in the blanks. Use: will use, will be connected, you will refer, if you want, will be. Use Capital letter where necessary.

1. If I link all the applications, there _____ many point-to-point connections.
2. If various components or services need to interact, they _____ to a message bus.
3. _____ to a message bus architecture If you talk about a virtual machine to send messages.
4. If you want to have a less complex connection, you _____ a message bus architecture.
5. You will require a console client _____ to make sure that everything is OK with your bus



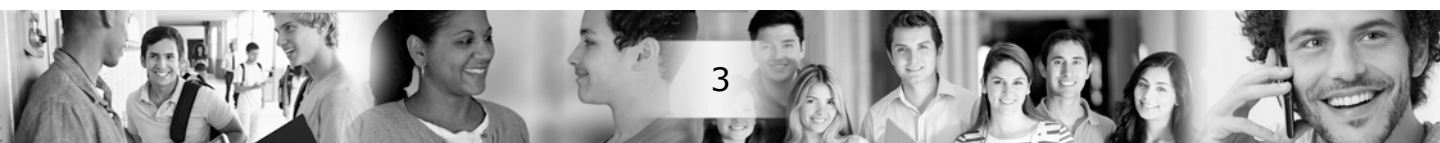
Let's Practice!

Message Bus Architecture and Its Possible Disadvantages

Instructions: Read the questions. Select the correct answers.

You have a desk where you give advice on systems-related matters. People come to you to ask you questions, and you have to provide good tips. What can you tell them?

1. Look, I understand that one of the advantages of message bus architecture is that it reduces the applications' complexity. Is there any drawback to it? (Hint: Parties need to have everything in common.)
 - a. One disadvantage may be that not all connected parties must follow the bus conventions.
 - b. One disadvantage may be that all connected parties must follow the bus conventions.





Let's Practice!

2. One of the biggest points in message bus architecture is that modifiability is more practical. Are there any possible problems related to this? (Hint: Bus interface is not as strong as it may seem.)
 - a. The bus interface may be delicate, so if modifiability lowers, it'll break down.
 - b. The bus interface is really strong, but sometimes the modifiability lowers because of unexpected reasons.
3. I've heard that dependency on a bus or central cable makes operation easier. Are there any risks related to this? (Hint: Problems in the central cable generate operation issues.)
 - a. One serious risk is that if the central cable encounters a problem, the whole network may break down.
 - b. There aren't many risks, really. If the main cable encounters some problem, the whole network can be fixed easily.
4. In a bus message architecture, intermediaries are removed, and I heard this is an advantage. Is there a disadvantage related to this? (Hint: Security reduces because of messaging.)
 - a. Not really. The security is kept, even if it sends messages indiscriminately.
 - b. One possible drawback is that the security is lowered since it sends messages indiscriminately.
 - c. I don't think so. The security is not lowered because it doesn't send messages indiscriminately.
5. I was told bus message architecture lowers costs, but my system has many devices. Any problem with this? (Hint: With more connections, efficiency may reduce.)
 - a. Not really. The number of devices does not have an effect on it.
 - b. Efficiency is not going to lower with more devices, so there is no drawback with that.
 - c. Efficiency may be an issue as it will reduce if you have more devices.



Extension Activity

How can you explain this topic?

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for information about message bus architecture style.
- Write a definition for this concept.
- Think about three advantages and disadvantages for using this style.
- Make a mind map to present your ideas about this architecture style.
- Share your mind map with your partners.

Self-Assessment

Yes

No

- I search information about message bus architecture style.
- I write a definition about message bus architecture style.
- I say the advantages and disadvantages that a message bus architecture style has.
- I share my mind map with the class.

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Target Words

- **modifiability** (noun): the ability of making changes
e.g. It also provides modifiability just in case you need to add more components.
- **bulk** (noun): a big amount of something
e.g. You may use complementary shared memory techniques to manage a bulk of data.
- **bind** (verb): to link things together
e.g. Buses provide reliable messaging capabilities to clients as they bind services from different natures.
- **reliable** (adjective): trustworthy
e.g. Buses provide reliable messaging capabilities to clients as they bind services from different natures.
- **queue** (verb): to organize objects that are waiting in a line
e.g. A message bus doesn't have a limit in size and it queues the messages if necessary.



Learning Points

Conditional Sentences With If Clauses

You use **conditionals** with **if clauses** to express that a cause in the present has consequence in the future.

Conditionals have two sentences that express a cause, represented by the conjunction **if** plus an action in present, and a consequence which in this case is going to be represented by the future **will**.

The clause with **if** can be the first or the second sentence. Regardless the order, the meaning will be the same. Look at these two grammatical structures:

If + subject + **verb in present** + complement + (,) + subject + **will** + complement.

Example:

If we **use** the message bus architecture, we **will** connect diverse application at the same time.

Subject + **will** + complement + **if** + subject + **verb in present** + complement

Example:

We **will** connect diverse application at the same time **If** we **use** the message bus architecture.

Notice that in the previous example, we omitted the comma because the **if clause** was placed as the second sentence.

Examples

- *If I link all the applications, there will be many point-to-point connections.*
- *If various components or services need to interact, they will use a message bus.*
- *If you use a message bus, you will reduce complexity in application communication.*



Learning Points

Message Bus Architecture

This architecture style allows users to connect more than one application on a computing platform.

When we talk about the **message bus architecture style**, we must take into account that it is a software designed to bind computational, data, display, or graphic services, as well as program interpreters. All these services are able to communicate by using the bus; therefore, sharing data is a flexible process.

The word **bus** is used to resemble the hardware that connects various components in a computer. A message bus is a virtual machine with similar objectives. The main function of a bus is to distribute messages determined by a messaging protocol, and it ensures that such messages are delivered. There is not a limit on the size of the message and if it is necessary, the bus queues the messages for delivery. On the other hand, if only two servers need to communicate, the bus may allow a point-to-point connection.

Advantages	Disadvantages
The applications' complexity is reduced.	All connected parties must follow the bus conventions.
Modifiability is more practical.	If the bus interface is broken, the modifiability lowers.
There is a better performance since intermediaries are removed.	The security is lowered since it sends messages indiscriminately.
Scalability is improved.	The bus is the only point of failure.